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CELL ORGANELLES

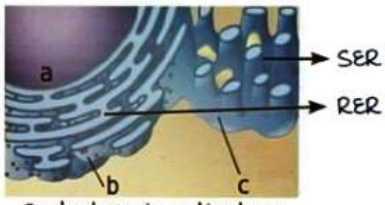
Endoplasmic Reticulum

00:00:30

Types

Rough endoplasmic reticulum (RER)

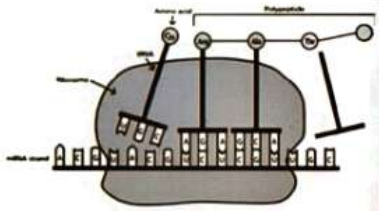
- Granular surface d/t presence of ribosomes.
- AKA granular ER.



Endoplasmic reticulum

Smooth endoplasmic reticulum (SER)

- Smooth surface d/t absence of ribosomes.
- AKA agranular ER.


Translation

ROUGH ENDOPLASMIC RETICULUM :

Functions :

1. Protein synthesis/translation.
2. Folding of proteins : Using chaperones/heat shock proteins (HSP).
3. Degradation of mis folded proteins & quality control of proteins by unfolded protein response or ER stress response.

↑↑misfolded proteins → ER → 1. Halts protein synthesis.
2. Endoplasmic reticulum associated degradation (ERAD).

ERAD :

misfolded proteins are tagged to ubiquitin (kiss of death)

↓

Protein-ubiquitin complex

↓

moves to proteasome (Graveyard of the cell)

↓

Degradation

Applied aspect : misfolded protein diseases.

1. Alzheimer's disease : more common.
   - Due to accumulation of misfolded Aβ amyloid protein.
2. Prion disease :
   - PrPc (Normal protein in CNS) → PrPsc (Prion protein : misfolded).
   - Eg : Creutzfeldt Jakob disease (very fatal).

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SMOOTH ENDOPLASMIC RETICULUM :

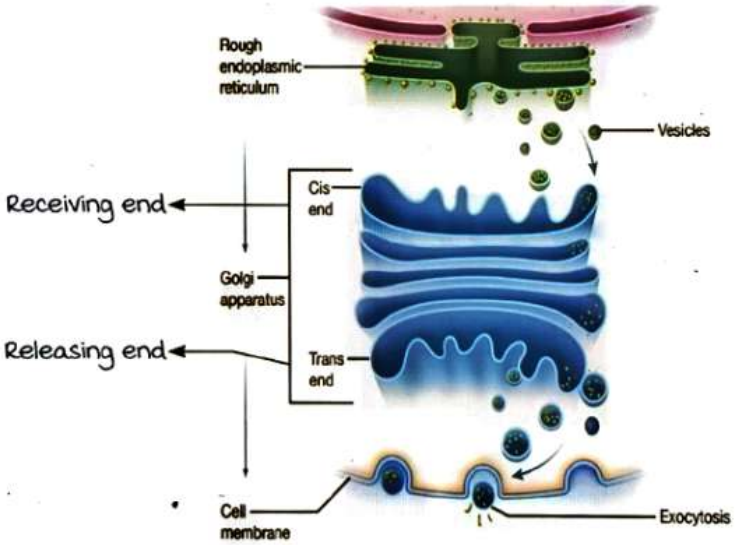
Functions :

1. Synthesis of steroid hormones :
   - Adrenal cortex : Aldosterone, cortisol.
   - Testis : Testosterone.
   - Ovary : Estrogen.
2. Storage of Ca^{2+} :
   - SER of skeletal muscle stores Ca^{2+} (Sarcoplasmic reticulum).
3. Drug detoxification (Xenobiosis) :
   - Major site : Liver SER.
   - Enzyme : Cytochrome P450 (CYP450).

Golgi Apparatus

00:14:20

Location : Close to RER.



Golgi apparatus

Functions :

1. Post translational modifications:

Proteins synthesised by RER
↓
Cis end of Golgi apparatus (GA)
↓
Glycosylation (Carbohydrate side chain is added to the protein)
↓
Glycoprotein (↑ stability of protein)
↓
Stored in vesicles
↓ Stimulus
Released from trans end

GA are abundant in secretory cells.
Eg : Antibody secreting plasma cells.
Goblet cells of GI tract.

2. Sorting of proteins :
   - GA determines where a protein is delivered.
   - Eg : Protein —Phosphotransferase→ Protein + mannose-6-phosphate
     ↓
     moves to lysosome
     ↓
     Degraded

Feedback

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Applied aspect :

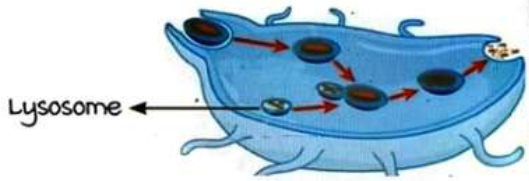
I cell disease :

Absence of phosphotransferase
↓
Mannose-6-phosphate not added
↓
Lysosomal function affected
↓
Proteins not degraded
↓
Inclusion bodies accumulate in cell

Lysosomes

00:23:30

Recycle bin of the cell.



Phagocytosis

Functions :

Acid mediated destruction.

- H^+ATPase acid hydrolase.
- Acid hydroxylase enzyme.

Note : Acid in the lysosome can destroy itself : Suicidal bags/residual bodies.

Autophagy :

Destruction of cell's own part.

Late stages of starvation
↓
Lysosomes engulf mitochondria
↓
Proteins released from mitochondria
↓
Undergo metabolism
↓
Energy for survival

Clinical application :

Neutrophil : High content of lysosomes.

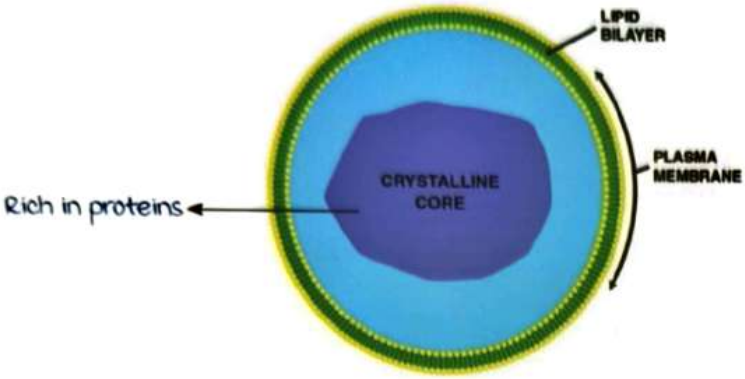
CHS1 gene —Normal→ LYST (Lysosomal transport protein)
↓ mutation
Chediak Higashi syndrome
↓
Patient is more prone to bacterial infections

Feedback

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Peroxisomes

00:31:55 ----- Active space -----



Peroxisomes/microbodies

Function :

1. Oxidation of very long chain fatty acids (VLCFA's).
   - Generate H_2O_2 (Free radical).
2. Degradation of H_2O_2 : By catalase.
   - Pseudocatalase : Commercial/drug form of catalase.
   - Used to treat free radical mediated disorders. Eg : Vitiligo.

Enzymes produced by peroxisomes :

1. Plasmalogens : Abundant in myelin sheath of nerve fibres.
2. Luciferase : Responsible for glow in fireflies.

Peroxisomal storage disorders :

Zellweger syndrome :

- PEX gene mutation.
- PEX codes for peroxins (Proteins for peroxisome synthesis).
- Features :
  - Accumulation of VLCFA's.
  - ↓ plasmalogens.

} Neurological impairment

Refsum's disease :

- PAHX gene mutation.
- Defect in phytanyl CoA α hydroxylase.
- Defective oxidation of phytanic acid.

Adrenoleukodystrophy :

- Defect in transport proteins.
- ↑↑ VLCFA's → Neurological impairment.

Feedback

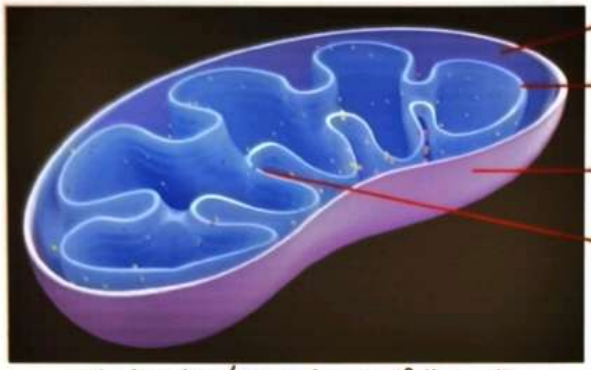
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Mitochondria

00:40:06



Intermembranous space

Inner membrane

Outer membrane

Cristae : Contains enzymes for oxidative phosphorylation

ATP synthesis

Mitochondria/power house of the cell

Mitochondrial DNA :

- Mitochondria has its own DNA (Endosymbiotic theory).
- Source : Ovum → Inheritance is always maternal.
- ds DNA with 16500 base pairs.
- 10 times more prone for mutation than nuclear DNA.
  - Repair of mutations in mitochondrial DNA is ineffective.
- Cannot function without nuclear DNA (To produce enzymes for ATP synthesis).

Mitochondrial disorders :

Affects organs with high metabolic requirements :

Features :

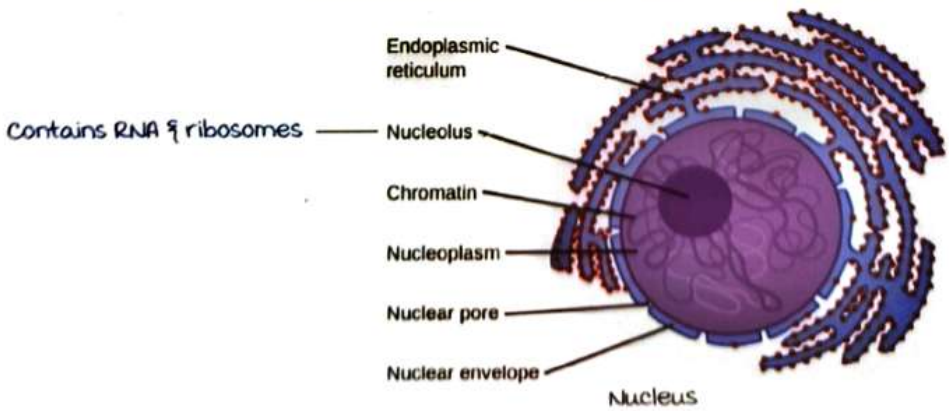
- Skeletal muscles : Contraction & relaxation is affected.
- CNS : Stroke like symptoms.
- GI tract is affected.
- ↑ susceptibility to infection.
- Lactic acidosis is common.

Mnemonic : MELAS.

- Mitochondrial Encephalopathy.
- Lactic Acidosis.
- Stroke.

Nucleus

00:48:13



Nucleus

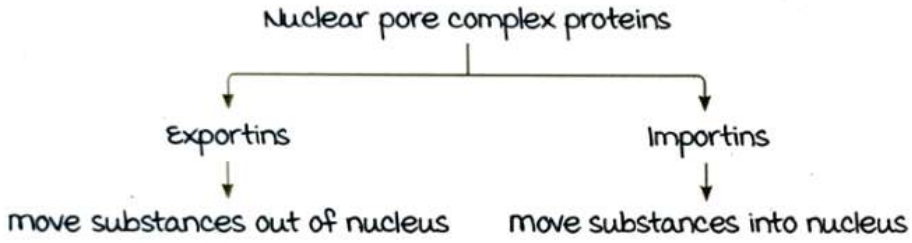
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Features :

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- Controls all cellular functions.
- Contain DNA (Chromosomes).
  - DNA + histones → Chromatin.
  - Structural repeating unit of chromatin : Nucleosome.
- Entry/exit of substances into the nucleus is regulated by Nuclear pore complex (NPC).
  - Substances expressing nuclear localization sequence (NLS) can move across NPC.



Marker Enzymes

00:52:30

Helps identify cell organelles.

Cell organelle	Marker enzyme
Cell membrane	Na^+ -K^+-ATPase Adenylyl cyclase
Endoplasmic reticulum	Glucose-6-phosphatase
Golgi apparatus	Galactosyl transferase
Lysosomes	Acid phosphatase
Peroxisomes	Catalase Urate oxidase
mitochondria	ATP synthase Creatine Kinase
Nucleus	RNA polymerase Histone deacetylase

Feedback